

HEALTHCONNECT CLINIC EXPERIENCE LAB

Project Title: Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

EXECUTIVE SUMMARY

Week 6 builds on the Week 5 findings by moving from descriptive analysis to advanced analysis, validation, segmentation and decision support. The main objective was to determine whether the strongest Week 5 patterns remained meaningful when multiple variables were analysed together.

The overall no-show rate remains 48.46%, compared with an attendance rate of 46.28%. The strongest risk indicators were previous no-shows and long booking lead time. Appointments booked 31+ days in advance had a 60.49% no-show rate, while the identified high-risk segment recorded a 61.02% no-show rate across 233 appointments.

Combining previous no-show history with booking lead time showed that patients with repeated previous no-shows and appointments booked far in advance represent particularly high-risk groups. Reminder analysis also showed lower no-show rates among patients who received reminders across all previous no-show groups. However, this represents an association and does not prove that reminders directly cause improved attendance.

Overall, Week 6 supports a shift from a general appointment-management strategy toward risk-based appointment support, where HealthConnect prioritises appointments with multiple risk characteristics.

ADVANCED ANALYSIS AND VALIDATED FINDINGS

PREVIOUS NO-SHOWS × BOOKING LEAD TIME

The combined analysis confirmed that previous no-show behaviour and booking lead time remain important risk indicators. Patients with repeated previous no-shows, especially those with 3+ previous no-shows, showed substantially higher no-show rates.

Risk becomes more concentrated when repeated previous no-shows are combined with long booking lead times, with some combinations reaching approximately 80% no-show rates. This suggests that risk should be assessed using combinations of factors rather than individual variables alone.

REMINDER EFFECTIVENESS

Reminder analysis showed lower no-show rates among patients who received reminders across every previous no-show group.

For example:

- **3+ previous no-shows:** approximately 76.00% without a reminder vs. 66.18% with a reminder.
- **2 previous no-shows:** approximately 66.67% vs. 55.65%.
- **1 previous no-show:** approximately 56.54% vs. 50.32%.
- **0 previous no-shows:** approximately 46.44% vs. 41.36%.

This indicates a consistent relationship between reminders and lower no-show rates. However, further statistical testing is required because the analysis does not establish causation.

LONG LEAD TIME × DISTANCE

Appointments booked 31+ days in advance remained high-risk across distance categories. No-show rates remained around or above 50% in several groups, with some higher rates among patients living farther from the clinic.

This suggests that long booking lead time remains an important risk characteristic even when distance is considered.

OTHER SEGMENTS

Appointment-day differences were relatively small. Sunday had the highest rate at approximately 50.47%, while Friday had the lowest at approximately 46.57%. However, Sunday appointments require caution because the Clinic Knowledge Base states that the clinic is closed on Sundays.

Appointment time also showed limited variation:

- Evening: **49.73%**
- Afternoon: **48.36%**
- Morning: **48.14%**

Age groups showed moderate differences, with the highest rate among ages 55–64 at 50.75% and the lowest among 65+ at 46.12%.

These variables provide useful context but are weaker differentiators than previous no-shows and booking lead time.

REFINED KPIS

Week 6 retained the main Week 5 KPIs and introduced additional decision-support measures:

KPI	Result
Total Appointments	5,000
Overall No-Show Rate	48.46%
Attendance Rate	46.28%
Cancellation Rate	5.26%
Reminder Coverage	72.68%
Long-Lead No-Show Rate	60.49%
High-Risk No-Show Rate	61.02%
High-Risk Appointments	233
Reminder No-Show Gap	4.03 percentage points

The new high-risk KPIs help management identify where targeted intervention may have the greatest potential value.

KEY BUSINESS INSIGHTS

Five major insights emerged from the Week 6 analysis:

1. **Previous no-shows are the strongest behavioural warning signal.** Patients with repeated previous no-shows have substantially higher future no-show risk.
2. **Long booking lead time is a major operational risk factor.** The 60.49% no-show rate for 31+ day appointments is considerably above the overall rate.
3. **Combined risk factors provide stronger decision support.** Patients with repeated previous no-shows and long booking lead times form particularly high-risk groups.
4. **Reminders are associated with lower no-show rates.** This relationship remains visible across previous no-show groups, although causation has not been established.
5. **Not all variables have equal decision value.** Previous no-shows, booking lead time and distance provide stronger signals than appointment day and appointment time.

BUSINESS RECOMMENDATIONS

HealthConnect should adopt a risk-based appointment support strategy.

- Provide additional appointment support to patients with 3+ previous no-shows, such as enhanced confirmation and follow-up communication.
- Monitor appointments booked 31+ days in advance because of their elevated no-show risk.
- Increase appropriate reminder coverage, particularly for high-risk appointments.
- Investigate whether reminder channels such as SMS, Email and WhatsApp produce different outcomes.
- Consider combining previous no-shows, booking lead time and distance into a future appointment-risk scoring system.
- Investigate transportation and accessibility barriers for patients living farther from the clinic.
- Investigate the relatively high no-show rate among follow-up appointments.
- Resolve the Sunday appointment data inconsistency before using Sunday patterns for operational decisions.

DASHBOARD IMPROVEMENTS

The Week 5 Power BI dashboard was enhanced to support advanced decision-making. New or refined visuals include:

- High-Risk No-Show Rate and High-Risk Appointment KPIs
- Previous No-Shows × Booking Lead Time risk analysis
- Reminder Effect by Previous No-Shows
- Long Lead Time × Distance analysis
- Monthly validation trend
- Appointment-day analysis
- Supporting age, appointment-time and appointment-type analysis

These improvements move the dashboard from basic reporting toward risk identification and decision support.

DATA SCIENCE CONTRIBUTION

Week 6 provides useful inputs for future predictive modelling. Potential features include: Previous no-shows, Booking lead time, Reminder status/channel, Distance to clinic, Appointment type, Age group, Appointment characteristics

The analysis also suggests that interaction effects may be important, particularly previous no-shows $\times$ booking lead time.

Before modelling, the Data Science track should assess missing values, class imbalance, feature quality, possible data leakage, interaction effects, train/test separation, model interpretability and fairness across patient groups.

The 233 high-risk appointments can also provide a useful basis for testing alternative risk-classification approaches.

LIMITATIONS

The analysis has several limitations. Missing values remain in distance, waiting time and reminder-channel fields. The Sunday appointment inconsistency has not yet been resolved. The analysis identifies associations rather than causation, meaning that lower no-show rates among reminded patients do not prove that reminders caused the improvement.

The high-risk segment should therefore be treated as a decision-support and prioritisation mechanism, not as a definitive prediction of individual patient behaviour. The dataset is also fictional and anonymised, so findings should not automatically be generalised to real healthcare populations.

WEEK 7 ANALYTICAL TESTING REQUIREMENTS

Week 7 should strengthen the Week 6 findings through formal testing. Key activities should include:

- Test whether the reminder/no-reminder difference is statistically significant.
- Test previous no-shows and booking lead time together.
- Investigate interaction effects such as previous no-shows $\times$ lead time and lead time $\times$ distance.
- Validate whether the 61.02% high-risk rate remains consistent under alternative risk definitions.
- Examine monthly trends for stability.
- Resolve the Sunday operating-day issue.

CONCLUSION

Week 6 successfully advances the HealthConnect Data Analytics project from descriptive reporting to advanced analytics and decision support. The analysis confirms that previous no-show behaviour and long booking lead time are the strongest risk indicators identified so far.

The 61.02% high-risk no-show rate, compared with the overall 48.46% rate, demonstrates the potential value of targeted risk-based support. Reminder analysis also shows a consistent association with lower no-show rates, although further testing is required to establish effectiveness.

The key Week 6 conclusion is that HealthConnect should move beyond analysing individual risk factors and begin focusing on combined risk profiles, particularly previous no-shows, booking lead time, reminders and accessibility-related factors.

These findings provide a stronger foundation for Week 7 statistical testing and predictive modelling, while maintaining appropriate caution around data quality, causation and patient-level interpretation.